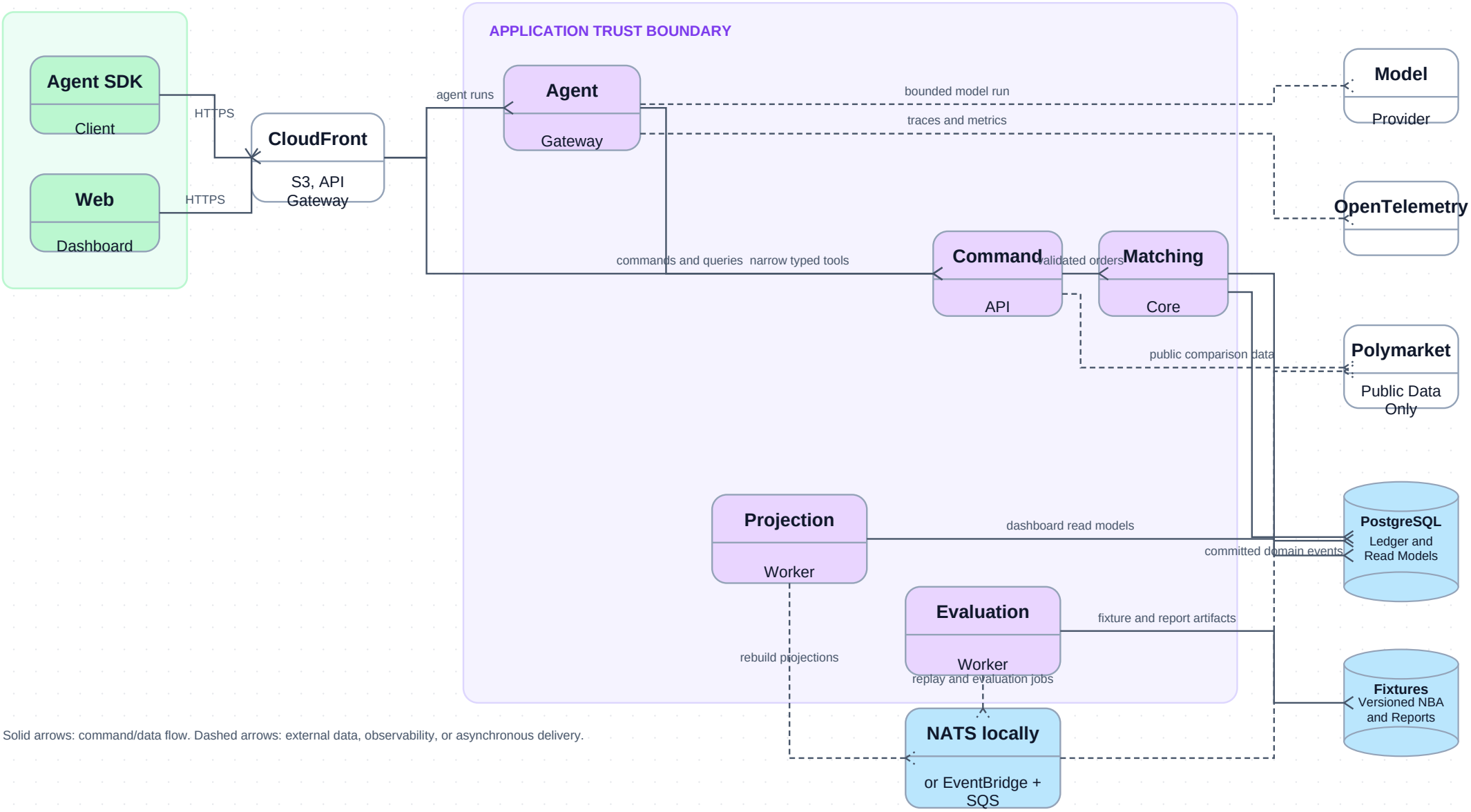
NBA Agent Market Sandbox - V1 System Architecture

Paper-only NBA prediction markets | agents request; deterministic services decide and record



Solid arrows: command/data flow. Dashed arrows: external data, observability, or asynchronous delivery.

Build order, guardrails, and demo story

Build this in sequence. Local correctness comes before cloud deployment or a real model call.

0	**Foundation** TypeScript monorepo, Compose, Postgres, NATS, test conventions, and CI. No cloud credentials.
1	**Correctness core** Fixture markets, deterministic matching, double-entry paper ledger, idempotency, and settlement.
2	**Database and events** Postgres transactions, market locks, transactional outbox, durable delivery, rebuildable projections.
3	**Human experience** Dashboard and API for market detail, order flow, portfolio, activity, and a rejected order.
4	**Constrained agent** Typed tools, policy checks, fake-model evaluations, then tightly capped real-model runs.
5	**Observability** Correlation IDs, traces, redaction, failure drills, replay, and evaluation reporting.
6	**AWS demo** CDK only after local parity: static web, Lambda/API Gateway, EventBridge/SQS, temporary RDS if needed.
7	**Optional comparison** Read-only Polymarket adapter. It never trades, signs, holds credentials, or settles simulator markets.

Normal operating mode

Local Docker/Testcontainers. AWS exists only for short public demos and release verification.

Budget rule

Alerts at $5, $10, and $20. $30 is the ceiling, never the normal monthly plan. Model calls have per-run and daily caps.

Avoid in V1

No NAT Gateway, EKS, MSK, Multi-AZ RDS, provisioned concurrency, permanent containers, or always-on ElastiCache.

Never in this project

No real-money trading, user funds, wallets, KYC, deposits, withdrawals, or external-feed settlement.